

Alexander Krett

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Education

Worcester Polytechnic Institute

Expected May 2028

M.S. in Computer Science, thesis-based

Framingham State University

May 2024

B.S. in Biochemistry, ACS-certified; Minor in Mathematics; *cum laude*

Research Experience

Subliminal Learning and Debiasing Research

Aug 2026–Present

Worcester Polytechnic Institute, Computer Science

Advisor: Dr. Walter Gerych; collaborator: Ronak Wani, Ph.D. student

- Built a reproducible Qwen2.5-7B-Instruct pipeline to study hidden preference transfer through generated number sequences, using LoRA fine-tuning and five paired random seeds across owl-preference and neutral conditions.
- Implemented layer-wise steering vectors and PyTorch hooks for WRING-inspired, norm-preserving rotations and projection baselines; investigate target-preference suppression, non-target drift, and language-model utility.
- Automated Slurm GPU workflows for generation, filtering, training, and evaluation, with resumable stages and structured experimental artifacts.
- Contributed a GLiNER biomedical entity-extraction prototype to MedRAG research, preserving source spans and confidence scores for concepts in clinical questions.

Data Analysis Internship and Honors Thesis

Sep 2023–May 2024

Framingham State University, Digital Humanities | Advisor: Dr. Kristen Bennett

- Developed five Python text-analysis workflows for the Kit Marlowe Project: word frequency, lexical dispersion, LDA topic modeling, word-adjacency networks, and word clouds.
- Published reproducible Jupyter notebooks and beginner-friendly Anaconda guides for Mac and Windows, enabling text analysis without command-line setup.

RNA-seq and Transcriptome Assembly Project

Jan–May 2023

Framingham State University, Biology | Advisor: Dr. Cheng-Chiang Wu

- Built a reproducible RNA-seq workflow using FastQC, Rcorrector, Trim Galore, Trinity, and BUSCO for read preprocessing, transcriptome assembly, and completeness assessment.
- Ran CentOS/Miniconda workflows with SGE on Boston University HPC resources at the Massachusetts Green High Performance Computing Center.

Scientific Software and Industry Experience

Novo Nordisk | Lexington, MA

Jun 2024–Aug 2026

Research Associate 1 (contract), Aug 2024–Aug 2026

Bio Analysis Intern, Jun–Aug 2024

- Developed Dockerized Python pipelines with a TypeScript/TSX interface for scientific data ingestion, processing, quality control, and reporting; reduced turnaround from approximately eight hours to fifteen minutes.
- Published automation to an internal application hub, extending standardized processing and QC workflows across R&D teams supporting nonclinical ADME and pharmacokinetic studies.
- Supported LC-MS/MS and HPLC-FLD assays for nonclinical ADME and pharmacokinetic studies.

Self-Hosted AI and Research Infrastructure

Independent Homelab

Approximately four years

- Developed Terraform configurations for Proxmox virtual machines and Ubuntu cloud-init templates, specifying compute, storage, networking, and full-clone provisioning.
- Built Ansible playbooks and reusable roles for host bootstrapping, Docker deployment, software updates, and service configuration; adapted community infrastructure patterns into a unified, version-controlled workflow.
- Configured containerized local AI serving with Ollama, IPEX-LLM, OpenVINO Model Server, and whisper.cpp, including Intel GPU and Vulkan backends for language and speech models.
- Built serial LLM benchmarks with warm-up exclusion and CPU/GPU resource sampling; implemented Whisper comparisons using Earnings-22 audio, word error rate, latency, and real-time factor.
- Automated SSH access controls, security updates, VM snapshots, NFS storage, service health checks, and notifications; integrated Ansible Vault with macOS Keychain for secret management.

Technical Skills

AI and data: Python, PyTorch, Hugging Face Transformers, PEFT/LoRA, GLiNER, pandas, NumPy, NLTK, matplotlib, Jupyter, R

Software and systems: TypeScript/TSX, Bash, Git, Docker, Linux, Ansible, Terraform, Proxmox, Slurm, SGE, LaTeX

Teaching, Leadership, and Community Engagement

President, FSU Outing Club

Sep 2022–May 2024

- Grew membership from 16 to over 130 students and annual funding from \$1,500 to \$14,000; built a full executive board and organized twice-weekly outdoor activities.
- Successfully advocated to eliminate participant fees, removing financial barriers; coordinated shared gear, transportation, and vendors for climbing trips and hikes.

Operations Manager, FSU Student Activities Board

Sep 2023–May 2024

Coordinated campus event logistics and programming within a \$165,000 annual budget.

Teaching and Academic Engagement

Guest lecturer, digital humanities, FSU; Organic Chemistry II tutor, FSU (Jan–May 2022).

Member, Data Science AI Council Club, WPI

Sep 2026–Present

Assist with planning and organizing club events.

Selected Presentations and Awards

Presentations

Kit Marlowe Project poster, inaugural Center for Digital Humanities launch, FSU (May 2024).

Enzyme-inhibition capstone poster, Biochemistry, Chemistry, and Food Science Department, FSU (2023).

Flame-resistant fabric calorimetry research, Massachusetts Undergraduate Research Conference, virtual presentation (2022).

Framingham State University awards

Kathleen Ryan Roberts Advocacy Award (2024); Paul T. Murphy University and Community Service Award (2023); Excellent E-Board Member of the Year (2023); Organic Chemistry Achievement Award (2021).

Additional Professional Experience

Voyager Therapeutics, Cambridge, MA: Laboratory Operations Manager Assistant (contract), Aug 2020–Aug 2021; Laboratory Technician, Aug 2019–Aug 2020. Supported lab planning and supply forecasting during COVID-19.